

Mark schemes

Q1.

- (a) Distance of particle from the origin is $\{(4\cos 3t)^2 + (4\sin 3t)^2\}^{\frac{1}{2}}$
- M1

= 4 which is a constant
 $\therefore$ particle is moving in a circle centre the origin

A1
2

- (b) $\mathbf{v} = \frac{d\mathbf{r}}{dt}$

$$\mathbf{v} = -12 \sin 3t \mathbf{i} - 12 \cos 3t \mathbf{j}$$

M1 for either term correct

M1A1
2

- (c) $\mathbf{a} = \frac{d\mathbf{v}}{dt}$

$$\mathbf{a} = -36 \cos 3t \mathbf{i} + 36 \sin 3t \mathbf{j}$$

M1 for either term correct

M1A1
2

- (d) $\mathbf{a} = -9 (4 \cos 3t \mathbf{i} - 4 \sin 3t \mathbf{j})$
 $= -9 \mathbf{r}$
 $k = -9$

B1 for 9

B2
2

- (e) Acceleration is towards centre of circle (or origin)

E1
1

[9]

Q2.

- (a) $\mathbf{v} = \frac{d\mathbf{r}}{dt}$

$$\mathbf{v} = -2 \sin \frac{1}{4}t \mathbf{i} - 2 \cos \frac{1}{4}t \mathbf{j}$$

No i, j: no marks

M1 A1

2

(b) Speed is $\{(-2 \sin \frac{1}{4}t)^2 + (-2 \cos \frac{1}{4}t)^2\}^{\frac{1}{2}}$

M1

$$= 2 \left(\sin^2 \frac{1}{4}t + \cos^2 \frac{1}{4}t \right)^{\frac{1}{2}}$$

clear use of $\sin^2 \theta + \cos^2 \theta = 1$

m1

= 2 which is a constant

Use of 2 values SC1

A1

3

(c) Magnitude of **r** is

$$\{(8 \cos \frac{1}{4}t)^2 + (8 \sin \frac{1}{4}t)^2\}^{\frac{1}{2}}$$

$\mathbf{a} = -k\mathbf{r} \Rightarrow \text{circle}$

SC2

M1

= 8 which is a constant

$\therefore$ Particle is moving in a circle

A1

2

(d) Using $v = a\omega$

M1 for their $\frac{b}{c}$ if both found

M1

Angular speed is 0.25

A1

2

(e) $\mathbf{a} = -\frac{1}{2} \cos \frac{1}{4}t \mathbf{i} + \frac{1}{2} \sin \frac{1}{4}t \mathbf{j}$

M1 A1

2

(f) Magnitude of acceleration is $\frac{1}{2}$



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Examiner reports

Q1.

In part (a) many candidates attempted to prove that the particle moved in a circle by showing that the particle was a constant distance, 4, from the origin at certain times. The general proof that the particle was always 4 from the origin was required. In parts (b) and (c) a small number of candidates did not include the **i** and **j** in their solutions. In part (d), many candidates attempted to divide vector **a** by vector **r** to obtain the value of **k**. In spite of the fact that two vectors cannot be divided in this way, many candidates found **k** to be -9. In part (e) a significant proportion of candidates assumed that the vectors **i** and **j** were in compass directions and gave the acceleration as, for example, north west.

Q2.

This question was badly answered. Many could not differentiate $\sin \frac{1}{4}t$ correctly, with $\cos \frac{1}{4}t$, $\cos \frac{1}{4}$ and $\frac{1}{4}t \cos \frac{1}{4}t$ all being seen regularly. Many of those who did find **v** correctly struggled to prove that $|\mathbf{v}|$ was a constant: they knew that $\sin^2 \theta + \cos^2 \theta = 1$, but often $(2 \sin \frac{1}{4}t)^2 + (2 \cos \frac{1}{4}t)^2$ became $4 + 4 = 8$.

Many dubious attempts were made to prove that the particle was moving in a circle. Only a few used the simple method: $r = 8$. In part (d), $v = \omega r$ and $v = \frac{\omega^2}{r}$ were used in equal measure. Candidates who used $v = \frac{\omega^2}{r}$ were misquoting the formula $a = \frac{\omega^2}{r}$. Often the values of **r** and **v** which candidates substituted were in vector form, with random attempts made at the division of two vectors.

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